Applied Forecasting of Financial Data

Presentation

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What is Forecasting?

Forecasting has fascinated humanity for thousands of years. At various points in history, attempts to predict the future were viewed as divine insight, while in other contexts they were treated with suspicion or even criminalized. It has been used as a means to predict the weather, natural disasters, and commonly in the modern era the price of commodities.

To put it simply: Forecasting is the process of predicting future trends, events, or conditions by analyzing historical data and current trends.

Project Overview and Goals

Goal: The acquisition, visualization, and forecasting of energy stock data for the sake of comparing traditional statistical forecasting, machine learning, and deep learning approaches for financial time series prediction.

Pipeline: Extraction $\rightarrow$ Preprocessing $\rightarrow$ Model Training $\rightarrow$ Evaluation

Forecasting Use Cases

- Pattern Recognition
- Informed decision making
- Risk Management
- Identifying opportunity

Stocks Selected and Scope of our Analysis

Traditional Energy

[XOM] Exxon Mobil

[CVX] Chevron

- Strong correlation with inflation and oil prices
- Extensive historical market data availability

Upstream Exploration

[COP] ConocoPhillips

- Strong sensitivity to crude oil prices
- Representative upstream energy producer

Infrastructure & Services

[BKR] Baker Hughes

- Captures industrial and supply chain exposure
- Broader energy sector representation

Renewable Energy

[RUN] Sunrun

- Renewable energy benchmark
- Contrast to traditional oil and gas behavior

The Methodologies we explored

Statistical

ARIMA
SARIMA

- Classical time series forecasting
- Models temporal autocorrelation
- SARIMA extends ARIMA with seasonality
- Strong statistical baseline

Machine Learning

XGBoost

- Gradient boosting framework
- Feature engineering driven
- Captures nonlinear relationships
- Uses lagged prices, returns, volume, price spreads

Deep Learning

LSTM

- Recurrent neural network architecture
- Designed for sequential data
- Learns long-term temporal dependencies
- Reduces manual feature engineering

ARIMA

AR: Auto Regression

I: Integrated

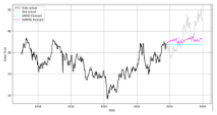
MA: Moving Average

SARIMA

S: [Seasonal] + ARIMA

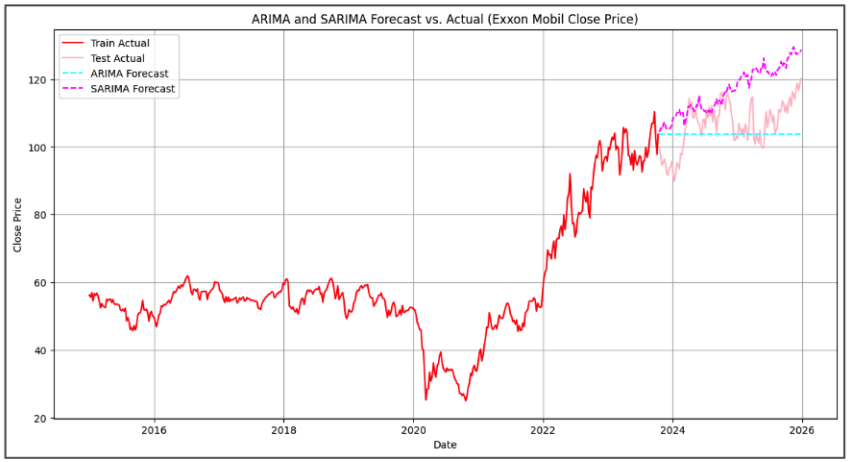
- ARIMA and SARIMA are statistical models used to forecast time series data (data recorded over time).
- In our case, the data consists of daily closing prices from five energy sector stocks.
- Both models were trained on historical stock market data from 2015 to 2023.

ARIMA and SARIMA Results

TICKERS **(OUR STOCKS)**	**RUN** Sunrun 	**CVX** Chevron 	**COP** ConocoPhillips 	**XOM** ExxonMobil 	**BKR** Baker Hughes 
ARIMA	Flat ~$15	Flat ~$152	Flat ~$118	Flat ~$104	Flat ~$34
SARIMA	Oscillated ~$15-23.	Trended up to ~$178.	Projected up to ~$140.	Propped all the way up to ~$128.	Modest growth to ~$36.
PERFORMANCE	Sarima is better, but both ignored the 2021 spike entirely.	Sarima > Arima. A little too optimistic from actual.	Both models missed, COP reversed in actuality.	Sarima performed well, where XOM's steady trend matched the forecast.	Big miss for both, BKR's unpredictable growth was not noticed by either model.

ARIMA and SARIMA Results

Best / Worst

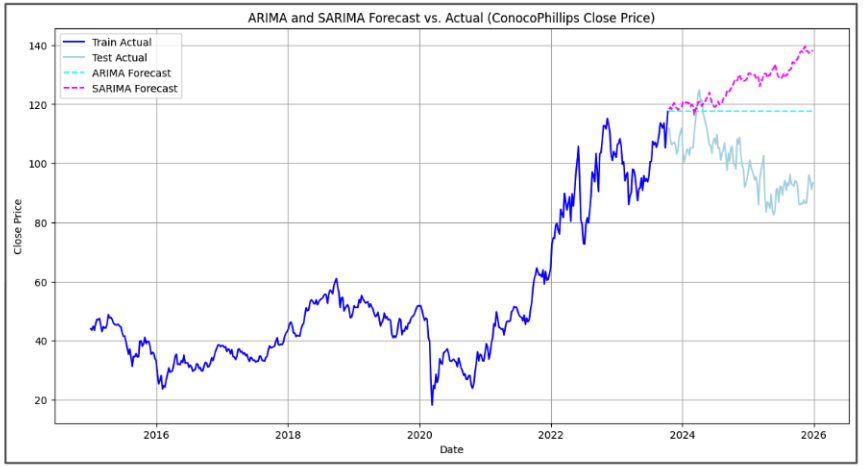


Best Case (XOM)

SARIMA projected continued upward movement. Exxon's steady trend matched closely.

Worst Case (COP)

SARIMA predicted growth to ~$140. The stock reversed and dropped to ~$85–95.

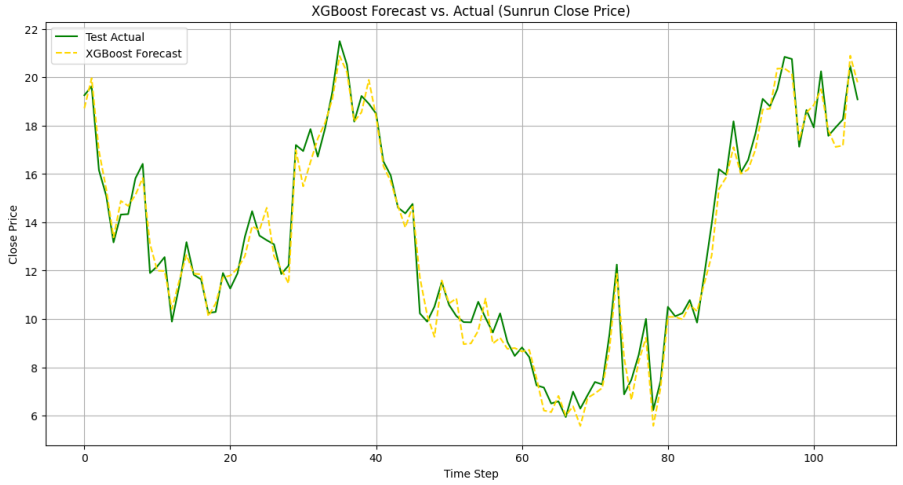


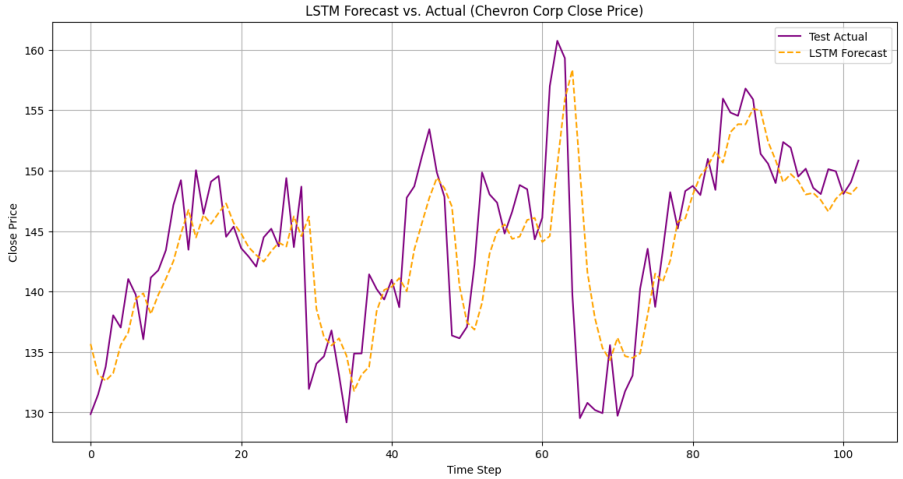
XGBoost Model

- **Ensemble Learning** – Machine learning approach which uses gradient boosting to sequentially correct errors from previous decision trees.
- **Implementation** – Converts time series data into engineered features (lag features, moving averages, technical indicators).
- **Strengths** – Better at capturing nonlinear relationships than ARIMA. Built in Lasso-Ridge regularization helps prevent overfitting by penalizing larger coefficients.
- **Limitations** – Heavily feature engineering dependent, weaker long-term forecasting capabilities, does not inherently model temporal structure.

XGBoost Results

- **Predictions** – Overall performance was strong, particularly for Sunrun and Chevron.
- **Features** – High Price was consistently ranked as the most important feature by a large margin, generally followed by Low Price, Open–Close Difference, Return, and Volume.



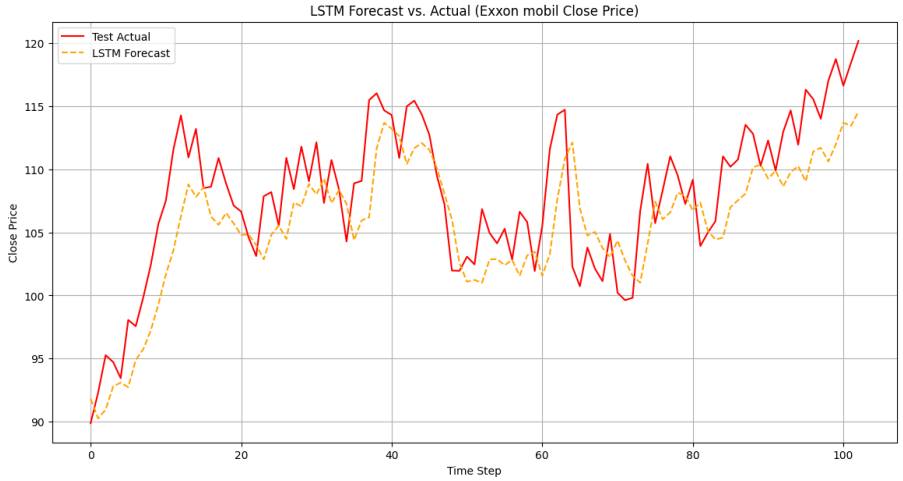


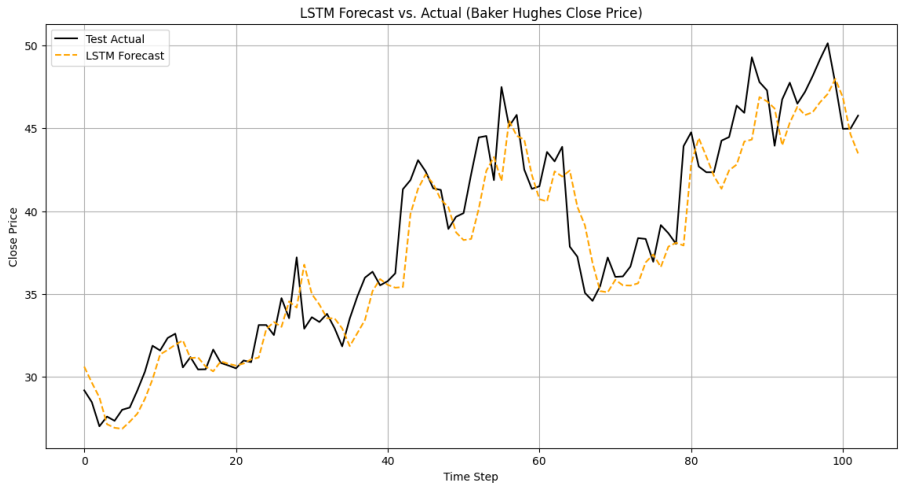
LSTM Model

- **Deep Learning Approach** – A Recurrent Neural Network (RNN) designed to learn long-term sequential patterns; avoids the "short memory" problem of traditional models.
- **Implementation** – Applied Min-Max Scaling (0–1), built a 60-day sliding time-window to predict next-day closing price, added Dropout layers to reduce overfitting.
- **Weaknesses** – Computationally expensive, harder to interpret, and still struggles with sudden external shocks.

LSTM Results

- **Predictions** – Strongest long-term performance for BKR and XOM; successfully tracked upward momentum where ARIMA flatlined.
- **Where It Excelled** – Captured sequential trend direction with high accuracy over extended forecasting horizons.
- **Where It Struggled** – Under-predicted extreme sudden price spikes in BKR driven by external market factors.

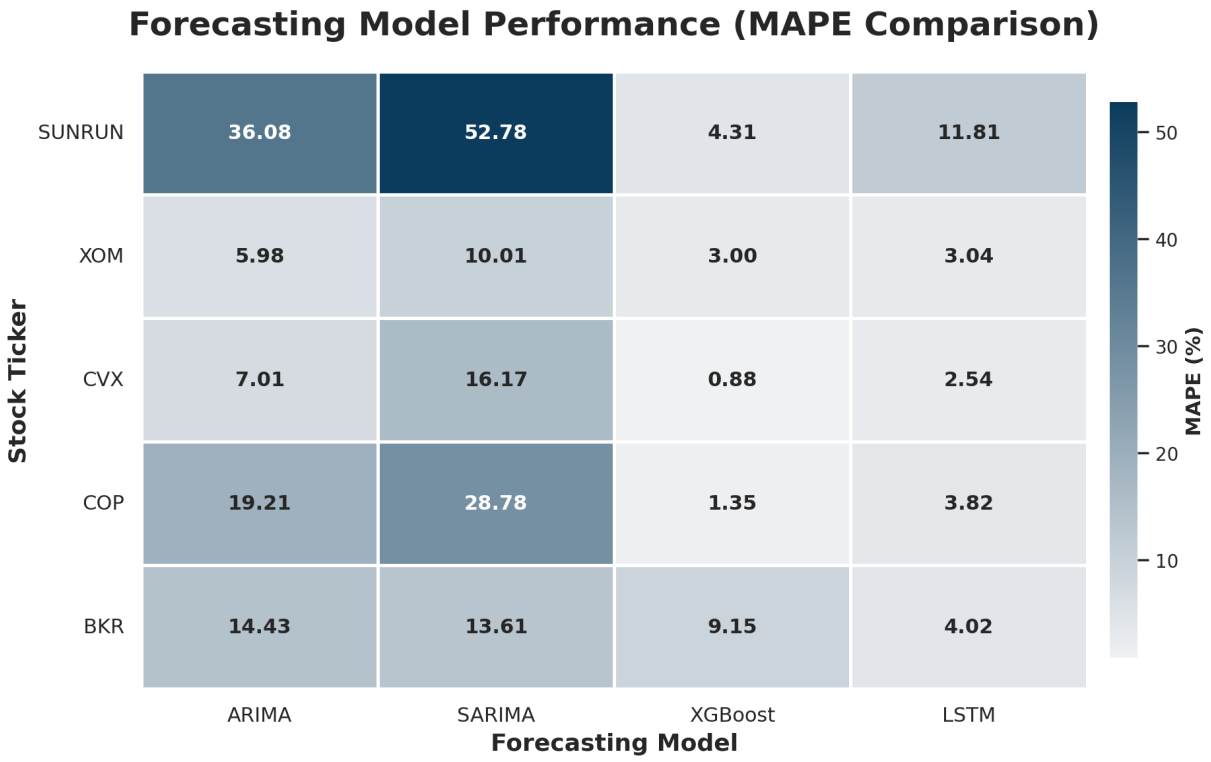




Raw Metrics Comparison Across All Tickers

Ticker	Model	MSE	MAE	RMSE	MAPE (%)
SUNRUN	ARIMA	19.34	3.86	4.40	36.08
	SARIMA	43.93	5.30	6.63	52.78
	XGBoost	0.40	0.52	0.63	4.31
	LSTM	3.28	1.39	1.81	11.81
XOM	ARIMA	56.76	6.40	7.53	5.98
	SARIMA	147.46	10.42	12.14	10.01
	XGBoost	21.55	3.34	4.64	3.00
	LSTM	16.08	3.31	4.01	3.04
CVX	ARIMA	141.57	9.60	11.90	7.01
	SARIMA	575.05	22.74	23.98	16.17
	XGBoost	2.71	1.26	1.65	0.88
	LSTM	25.81	3.61	5.08	2.54
COP	ARIMA	406.98	18.17	20.17	19.21
	SARIMA	949.09	27.11	30.81	28.78
	XGBoost	5.01	1.40	2.24	1.35
	LSTM	20.97	3.67	4.58	3.82
BKR	ARIMA	55.25	5.87	7.43	14.43
	SARIMA	40.81	5.29	6.39	13.61
	XGBoost	35.44	4.01	5.95	9.15
	LSTM	4.27	1.58	2.07	4.02

Error Distribution Across Models



- XGBoost achieved the lowest median forecasting error.
- LSTM also performed strongly with relatively low variance.
- ARIMA and SARIMA showed larger error distributions.
- Seasonal structure did not improve forecasting accuracy.

Conclusion

- XGBoost produced the strongest forecasting performance for Chevron and Sunrun.
- LSTM performed best for XOM and BKR by capturing long-term sequential patterns more effectively.
- ARIMA and SARIMA captured overall trends but generally struggled during volatile market periods.
- Different stocks responded better to different forecasting approaches depending on their market behavior and volatility.
- The results showed that no single model performed best for every stock.